Technovision 2026-27

Technical Club “FLUX”

Rulebook for RoboRumble:

Participation rules:

1. Team size should not exceed 4 members. Individual registrations are also allowed.
2. All members in a team should be enrolled in the same institution.

Competition setup:

1. The participants are required to construct a bot capable of completing the track in the allotted time.
2. The track will consist of many hurdles which the participants must overcome/complete.
3. Score will be calculated on completion of the track. Finishing the track is mandatory.

Competition rules:

1. No person can join more than one team .
2. The winners will be decided on basis of highest score and minimum time.
3. Every team will be allowed one run only.
4. A team may restart from the previous checkpoint at any point in the run.
5. After failing to complete an obstacle 4 times, the current obstacle will be skipped, and add 20 second time penalty will be added.
6. After usage of technical timeout, the run will continue from the last checkpoint.
7. Each participant will be allowed to skip one obstacle.

Bot rules:

1. Weight of the bot should not exceed 3 Kilograms.
2. The bot must be able to fit within a Hidden equation volume. An error of up to Hidden equation for each dimension will be tolerated.
3. Voltage difference between no two points in the bot should be greater than 12 V.
4. Intentionally damage to the track will be grounds for disqualification.

Marking rules:

1. Taking a timeout will incur a Hidden equation-second addition to the total time taken.
2. The final score will be the calculated with the following formula:

Total Score = (540 – TU – (MNP*3) - (MJP*15))

1. TU = Time Used
2. MNP = Minor Penalties
3. MJP = Major Penalties

Venue rules:

1. It is expected of participants to maintain the decorum of the venue.
2. Violent outbreaks, non-professional behavior towards other parties will be cause for disqualification and removal from the venue.

Technovision 2026-27

Technical Club "FLUX"

Official Rulebook: RoboWalk

Team Composition & Eligibility Rules

1. Team Size: A team may consist of a minimum of 2 member and a maximum of 4 members.
2. Institution Rules: All members of a team must be enrolled students of the same l institution.
3. Single Entry: No participant can be a member of more than one team in this competition.

Bot Specifications (No Circular Wheels Allowed)

1. The bot must not use standard circular wheels, rollers.
2. Any mechanism relying on rolling continuous wheel surfaces for primary motion will result in immediate disqualification.
3. Allowed mechanisms: You can use legs, mechanical joints that walk like legs, or crawling and stepping feet to move your robot.

Dimensions & Volume:

1. The bot must fit within a maximum bounding box of 30 cm × 30 cm × 30 cm at the start of the race.
2. A tolerance of up to 5% on each dimension is allowed.

Weight Limit:

1. The total weight of the bot (including power supply/batteries) must not exceed 3.0 kg.
2. The potential difference (voltage) between any two points on the robot must not exceed 12 V DC.

Control Mechanism:

1. The bot may be **wired** or **wireless** (RF, Bluetooth, or autonomous).
2. For wired bots, Keep the wire loose at all times. Pulling or dragging the robot with the wire will result in a penalty.

Track Specifications & Race Setup

1. The race track consists of a designated lane with a Start Line, Checkpoints, and a Finish Line.

2. The terrain may feature varying surfaces (e.g., smooth acrylic/wood, mild rubber mats, low-profile speed bumps, or slight inclines).
3. Finishing the track from Start to Finish is mandatory to qualify for ranking.

Competition & Track Rules

1. **Restart & Checkpoints:**
    - If your robot flips, gets stuck, or breaks down, you can restart from the last cleared checkpoint.
    - A manual intervention to reset the bot will incur a time penalty.
2. **Technical Timeout:**
    - Each team is allowed **one technical timeout** of up to **2 minutes** for quick repairs or battery replacements during their run.
    - Taking a technical timeout incurs a **10-second penalty** added to the final time.
    - After the timeout, the run resumes from the last cleared checkpoint.

Marking Scheme & Scoring

Winners will be determined based on the Highest Final Score (or Lowest Effective Time).

A. Final Score Formula

Hidden equation

Hidden equation

Where:

- Hidden equation **(Time Used):** Total elapsed time taken to clear the track in seconds.
- Hidden equation **(Minor Penalties):** Count of minor infractions (e.g., touching track walls, line step-out, wire pulling).
- Hidden equation **(Major Penalties):** Count of major infractions (e.g., manual resets, technical timeouts, track resets).

Venue Rules & Conduct

1. **Decorum:** All participants are expected to maintain professionalism and sportsmanship at all times within the venue.
2. **Disqualification Grounds:**

- Any violent behavior, ungentlemanly conduct, or abusive language toward judges, volunteers, or opponent teams will lead to **immediate disqualification and removal from the venue**.
- Intentional damage to the competition track, timing equipment, or venue property will result in disqualification and potential financial penalties.

3. **Judges' Authority:** The decision of the track judges and event coordinators is final and binding in all circumstances.